

PRODUCT DOCUMENTATION



SolarPro Platform

A full-stack solar sales and client management platform — Flutter mobile/desktop app, Python FastAPI backend, MongoDB database, and a public-facing landing page. Built for solar selling companies to manage clients, quotations, installations, and payments end-to-end.

Flutter (Android · iOS · Windows)

Python FastAPI

MongoDB

Hostinger VPS

Admin + Client Portals

v1.0

INFRASTRUCTURE

Tech Stack



Flutter + Dart

Cross-platform frontend



Python + FastAPI

REST API backend



MongoDB

Primary database



Hostinger VPS

Backend + landing page host



Landing Page

HTML/CSS hosted on VPS



JWT Auth

Token-based security

SYSTEM DESIGN

System Architecture

The Flutter app targets Android, iOS, and Windows from a single Dart codebase. All data is securely fetched from a Python FastAPI backend hosted on Hostinger VPS. A public-facing landing page is served from the same server and provides download links for each platform. Data is stored in the company's existing MongoDB database.

Flutter Application

A single Flutter codebase compiles to three platforms. The app has two distinct portals — Admin and Client — accessible after role-based login.



Android

APK / Play Store distribution.
Minimum SDK 21+.



iOS

IPA / App Store distribution.
iOS 13+ support.



Windows

Desktop installer (MSIX). For
admin-heavy use.

Shared App Features

- ✓ Role-based login (Admin / Client) with JWT token auth
- ✓ Push notifications (FCM) for quotes, updates, and alerts
- ✓ Offline-capable screens with local caching
- ✓ Dark / Light theme support
- ✓ Multi-language support (English + regional languages)
- ✓ In-app document viewer (PDF quotes, invoices)
- ✓ Secure token storage using flutter_secure_storage

Admin Panel Features

The admin portal is used by the solar company's internal team — sales reps, managers, and support staff.



Dashboard & Analytics

- ✓ Overview cards: total clients, active leads, revenue, installations
- ✓ Monthly revenue chart and sales funnel visualization
- ✓ Pending tasks and follow-up reminders
- ✓ Top-performing products / packages

Client Management (CRM)

- ✓ Add, edit, and manage client profiles
- ✓ Assign clients to sales representatives
- ✓ Track lead status: New → Contacted → Quoted → Won / Lost
- ✓ Notes and activity timeline per client
- ✓ Search, filter, and sort clients by status, region, package

Product & Package Management

- ✓ Add and manage solar panel packages (1kW, 3kW, 5kW, 10kW, custom)
- ✓ Set pricing, components (panels, inverters, batteries), and availability
- ✓ Upload product images and specification sheets
- ✓ Mark packages as active, discontinued, or promotional

Quotation Management

- ✓ Generate professional PDF quotations for clients
- ✓ Include site survey data, package details, pricing breakdown
- ✓ Send quotes directly to client via app notification + email
- ✓ Track quote status: Draft → Sent → Accepted / Rejected
- ✓ Revision history for each quotation

Installation & Order Tracking

- ✓ Create installation orders upon quote acceptance
- ✓ Assign installation teams and schedule dates
- ✓ Track installation stages: Scheduled → In Progress → Completed
- ✓ Upload site photos and installation reports
- ✓ Generate completion certificates

Payment & Finance

✓ Record payments (advance, milestone, final)

✓ Generate GST-compliant invoices

✓ Track outstanding dues per client

✓ Payment receipts shared with clients via app

✓ Financial summary reports (monthly/yearly)

Team & User Management

✓ Create admin sub-accounts with role-based permissions (Super Admin, Sales Manager, Sales Rep, Support)

✓ Assign client portfolios to team members

✓ Activity logs and audit trails

Notifications & Communication

✓ Send push notifications to individual clients or broadcast to all

✓ In-app messaging with clients

✓ Automated reminders for follow-ups, payment dues, and site visits

CLIENT SIDE

Client App Features

The client portal is what the end-customer (solar buyer) sees after downloading the app and being onboarded by the sales team.

Client Dashboard

✓ Summary of current order/installation status

✓ Active package details and system specifications

✓ Upcoming appointment or site visit dates

✓ Quick actions: View Quote, Track Installation, Pay Now, Contact Support

My Quotations

✓ View all quotations received from the company

✓ Download quotes as PDF

✓ Accept or request revision on a quote

✓ Compare multiple quote versions side by side

Order & Installation Tracking

✓ Real-time installation status with step-by-step progress tracker

✓ View assigned installation team contact details

✓ Scheduled visit dates and reminders

✓ Post-installation checklist and sign-off

Payments

✓ View payment schedule and upcoming dues

✓ Payment history with downloadable receipts

✓ Online payment (UPI, Net Banking, Card) integration

✓ Invoice download (GST compliant)

Solar Savings Calculator

✓ Interactive calculator — input monthly electricity bill

✓ Output: estimated savings, payback period, CO₂ offset

✓ Units generated per day/month based on installed capacity

✓ Comparison graph: electricity bill before vs. after solar

Support & Communication

✓ In-app chat with assigned sales/support representative

✓ Raise service requests or complaints

✓ View status of open support tickets

✓ FAQ and knowledge base section

Documents & Certificates

- ✓ Store and access all documents: agreement, invoice, warranty, subsidies
- ✓ Download completion and warranty certificates
- ✓ Government subsidy application status tracking

Notifications

- ✓ Push notifications for quote updates, installation stages, payment dues
- ✓ Custom notification preferences
- ✓ Notification history

SECURITY

Security

- ✓ Secure login with token-based authentication — sessions expire automatically for safety
- ✓ Role-based access control — Super Admin, Sales Manager, Sales Rep, and Client each see only what they're allowed to
- ✓ All data transmitted over HTTPS with SSL encryption
- ✓ Passwords are encrypted and never stored in plain text
- ✓ Protection against unauthorized access and abuse attempts
- ✓ Sensitive user data stored securely on the device
- ✓ Database access restricted to the application only